

Federated Exact Low-Rank Adaptation (FedEx-LoRA): Eliminating Aggregation Bias in Distributed Large Language Model Fine-Tuning

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ACL Link: <https://aclanthology.org/2025.acl-long.67/>

GitHub: <https://github.com/RaghavSinghal10/fedex-lora>

1. Title

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2. Problem Statement

Large Language Models (LLMs) such as LLaMA, Mistral, and GPT are increasingly deployed in privacy-sensitive domains — healthcare, legal services, and personalised e-commerce — where raw user data cannot be centralised. Federated Learning (FL) addresses this by training a shared model across distributed clients while keeping all data on-device. To reduce the computation and communication overhead of fine-tuning billion-parameter models, practitioners apply **Low-Rank Adaptation (LoRA)**: the pre-trained weight matrix W_0 is frozen, and only two low-rank matrices A and B are trained such that the weight update is:

$$\Delta W = B \times A, \text{ where } B \in \mathbb{R}^{(d \times r)}, A \in \mathbb{R}^{(r \times k)}, \text{ rank } r \ll \min(d, k)$$

However, combining FL with LoRA via standard **Federated Averaging (FedAvg)** introduces a critical mathematical flaw known as **Aggregation Bias** (also called decomposition drift). When the server independently averages the client adapters, it computes:

$$B_avg = (1/k) \times \sum B_i \text{ and } A_avg = (1/k) \times \sum A_i$$

The resulting global model update becomes $W_0 + B_avg \times A_avg$. However, this is mathematically incorrect:

$$W_ideal = W_0 + (1/k) \times \sum (B_i A_i) \neq W_0 + B_avg \times A_avg = W_FedAvg$$

The average of products is NOT equal to the product of averages. This systematic error degrades model accuracy, compounds across training rounds, and grows worse as the number of clients increases or as data becomes more heterogeneous (non-IID).

FedEx-LoRA solves this by computing a residual correction term ΔW_res at the server that captures the exact discrepancy:

$$\Delta W_res = (1/k) \times \sum (B_i A_i) - B_avg \times A_avg$$

The corrected global update then becomes:

$$W_0 + B_avg \times A_avg + \Delta W_res = W_0 + (1/k) \times \sum (B_i A_i) = W_ideal \checkmark$$

3. Motivation

3.1 Privacy and Data Sovereignty Regulations such as GDPR and HIPAA mandate that sensitive user data remain on-device. Federated fine-tuning is the only viable path to personalised AI in these domains. Any accuracy loss due to aggregation bias directly undermines the utility of these privacy-preserving systems. Exact aggregation is therefore not an academic concern — it is a prerequisite for production deployment.

3.2 Scale of Deployment LLMs with billions of parameters are deployed across thousands of edge devices. Even a small per-round bias in gradient aggregation compounds across training rounds, leading to significant accuracy degradation at scale. With 50 heterogeneous clients and 20 local epochs, the accumulated error from FedAvg-LoRA becomes clearly measurable as reduced task performance on standard benchmarks.

3.3 Core PDC Problem This is a textbook example of two classical Parallel and Distributed Computing challenges: (a) partial-state aggregation — how do we correctly reconstruct a global state from locally computed, non-commutative updates? and (b) the synchronisation-communication tradeoff — how does synchronisation frequency (local epochs)

affect accumulated drift and communication cost? FedEx-LoRA provides a mathematically principled answer to both.

3.4 Practical Industry Relevance Small businesses and independent sellers cannot afford centralised GPU clusters. Federated exact fine-tuning enables them to collaboratively train domain-specific LLMs — for product description generation, customer service, or ad copy — without sharing proprietary data. Exact aggregation ensures model quality justifies the privacy cost.

4. Objectives

1. Re-implement the FedEx-LoRA exact aggregation algorithm from scratch in PyTorch. *(Success metric: unit tests pass; residual error $\|W_{ideal} - W_{FedEx}\|_F < 1e-6$)*
 2. Simulate federated training on the GLUE NLU benchmark with non-IID data partitioned using Dirichlet $\text{Dir}(0.3)$. *(Success metric: match or exceed paper-reported accuracy on CoLA, RTE, MRPC, SST-2, QNLI, STS-B)*
 3. Quantify aggregation bias of FedAvg-LoRA versus FedEx-LoRA per communication round. *(Success metric: per-round divergence plot across $k \in \{10, 20, 50\}$ clients)*
 4. Measure communication overhead of the residual term across varying client counts. *(Success metric: bits-per-round comparison table: FedAvg-LoRA vs FedEx-LoRA vs FFA-LoRA)*
 5. Apply SVD-based compression to the residual to reduce communication cost. *(Success metric: accuracy drop $\leq 1.5\%$ at 50% bandwidth saving on GLUE SST-2)*
 6. Reproduce E2E NLG results with GPT-2 in federated settings. *(Success metric: BLEU, NIST, METEOR within 5% of paper-reported values)*
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5. Technical Contributions

C1 — Exact Residual Aggregation (Core Algorithm) The central contribution is the server-side residual correction. After receiving adapters $\{B_i, A_i\}$ from all k clients, the server computes B_{avg} and A_{avg} , then computes the residual $\Delta W_{res} = (1/k)\sum(B_i A_i) - B_{avg} \times A_{avg}$, and applies it to the global model. This guarantees mathematically exact weight reconstruction without transmitting full-rank weight matrices.

C2 — Non-IID Data Simulation via Dirichlet Partitioning Realistic heterogeneous federated environments are simulated using a Dirichlet distribution $\text{Dir}(\alpha)$ to partition the dataset across clients. We sweep $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$, where lower α produces highly skewed non-IID distributions, measuring how aggregation bias scales with data heterogeneity.

C3 — Frozen A-Matrix Strategy (FFA-LoRA Variant) A significant variant freezes matrix A globally after random initialisation and trains only B on each client. Since all clients share the same A , the residual ΔW_{res} becomes exactly zero, eliminating extra communication. This serves as a communication lower bound and important baseline.

C4 — SVD-Based Residual Compression (Novel Extension) Since ΔW_{res} can be high-rank (up to $k \times r$), we propose compressing it via truncated Singular Value Decomposition before transmission, retaining only the top r' singular components. The server transmits (U, Σ, V^T) of rank $r' \ll \text{rank}(\Delta W_{\text{res}})$. We sweep $r' \in \{r/4, r/2, 3r/4, r\}$ and plot the accuracy vs. communication cost Pareto frontier. This is a novel contribution beyond the base paper.

C5 — Synchronisation Frequency Analysis We empirically analyse how the number of local training epochs $E \in \{1, 3, 5, 10\}$ affects the Frobenius norm of ΔW_{res} , convergence speed in communication rounds, and final task accuracy. This maps directly to the classical PDC synchronisation frequency problem.

6. Datasets

Dataset 1 — GLUE Benchmark (Primary NLU Evaluation)

- Task: Natural Language Understanding
- Sub-tasks used: CoLA (grammar), RTE (entailment), MRPC (paraphrase), SST-2 (sentiment), QNLI (question inference), STS-B (semantic similarity)
- Splits: Standard train / validation
- Access: HuggingFace datasets library; gluebenchmark.com

Dataset 2 — MetaMathQA (Arithmetic Reasoning)

- Task: Multi-step arithmetic reasoning
- Details: 10,000-sample subset used for training Mistral-7B and Gemma-2 9B models; evaluated on GSM8K and MATH benchmarks
- Access: HuggingFace Hub (meta-math/MetaMathQA)

Dataset 3 — COMMONSENSE170K (Commonsense Reasoning)

- Task: Commonsense reasoning across 8 sub-datasets
- Sub-datasets: BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC-e, ARC-c, OBQA
- Models: Used with LLaMA series models
- Access: HuggingFace Hub and individual dataset repositories

Dataset 4 — E2E NLG Challenge (Natural Language Generation)

- Task: Structured data-to-text generation

- Details: Used to evaluate GPT-2 in federated settings with varying local training epochs
- Metrics: BLEU, NIST, METEOR, ROUGE-L, CIDEr
- Access: e2e-nlg.github.io; official challenge repository

All datasets are partitioned across clients using Dirichlet $\text{Dir}(\alpha)$ with $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ to simulate non-IID federated conditions.

7. Baseline Papers and Methods

B1 — FedAvg-LoRA (Primary Baseline / Performance Floor)

- Reference: McMahan et al., 2017 (FedAvg) + Hu et al., 2022 (LoRA, ICLR 2022)
- Description: Naive combination of Federated Averaging with LoRA. Independently averages B and A matrices, introducing aggregation bias. This is the direct method FedEx-LoRA improves upon. Expected to underperform FedEx-LoRA by 2–5% under non-IID conditions with 20+ clients.

B2 — FFA-LoRA (Frozen Feature Aggregation)

- Reference: Sun et al., 2024 (arXiv:2404.00218)
- Description: Freezes matrix A globally and trains only B on each client. The cross-term residual becomes zero, eliminating extra communication. Represents the communication-efficiency extreme but sacrifices model expressivity.

B3 — FlexLoRA (Heterogeneous Rank Adaptation)

- Reference: Wu et al., 2024
- Description: Allows each client to use a different LoRA rank based on hardware constraints. Uses SVD-based projection to a common rank before aggregation. Addresses hardware heterogeneity but still applies approximate aggregation.

B4 — RAVAN (Variance-Aligned Aggregation)

- Reference: Tripuraneni et al., 2024
- Description: Uses stochastic rounding and variance reduction to improve convergence under non-IID data. Represents the state-of-the-art in sophisticated approximate federated LoRA approaches.

B5 — Centralised LoRA (Upper Bound)

- Reference: Hu et al., 2022 (ICLR 2022)

- Description: Standard single-node LoRA fine-tuning with all data centralised. Represents the theoretical performance ceiling. FedEx-LoRA aims to close the gap between federated and centralised training to within 1–2%.
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8. Proposed Methodology / Flow

Pipeline Overview: Data Partition → Local Training → Transmit Adapters → Server Aggregation + Residual → Broadcast → Evaluate

Phase 1 — Environment Setup and Baseline Measurement

- Install PyTorch, HuggingFace Transformers, Flower federated framework, and PEFT library
- Implement FedAvg-LoRA baseline on GLUE SST-2 with $k = 10$ clients
- Quantify baseline aggregation bias by computing $\|W_{ideal} - W_{FedAvg}\|_F$ per communication round
- Apply Dirichlet $\text{Dir}(0.3)$ partitioning to simulate non-IID data distribution

Phase 2 — FedEx-LoRA Core Implementation

- Implement server-side exact residual: $\Delta W_{res} = (1/k)\sum(B_i A_i) - B_{avg} \times A_{avg}$
- Verify exactness: unit test $\|W_{ideal} - W_{FedEx}\|_F < 1e-6$ across all GLUE tasks
- Run federated training on all 6 GLUE sub-tasks; compare convergence with FedAvg-LoRA
- Scale to $k \in \{10, 20, 50\}$ clients to measure scalability

Phase 3 — Extended Evaluation

- Train on MetaMathQA (10K subset) for Mistral-7B; evaluate on GSM8K and MATH
- Evaluate on COMMONSENSE170K across all 8 sub-datasets with LLaMA models
- Fine-tune GPT-2 on E2E NLG; report BLEU, NIST, METEOR, ROUGE-L, CIDEr
- Vary local epochs $E \in \{1, 3, 5, 10\}$; measure effect on $\|\Delta W_{res}\|_F$ and final accuracy

Phase 4 — SVD Compression Extension (Novel)

- Compute truncated SVD of ΔW_{res} ; retain top r' singular values and vectors
 - Transmit $(U, \Sigma_{r'}, V^T)$ of rank r' instead of the full residual matrix
 - Sweep compression ratios $r' \in \{r/4, r/2, 3r/4, r\}$
 - Report accuracy vs. communication cost Pareto frontier for GLUE and E2E
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9. Expected Outcomes

Outcome 1 — Accuracy Recovery on GLUE FedEx-LoRA is expected to recover 95–100% of centralised LoRA accuracy on all 6 GLUE sub-tasks. FedAvg-LoRA is expected to lag by 2–5% under non-IID Dir(0.3) conditions with 20+ clients, directly validating the harm caused by aggregation bias.

Outcome 2 — Convergence Speed By eliminating aggregation bias, FedEx-LoRA should reach target validation loss in fewer communication rounds, improving overall training efficiency despite the additional residual term transmission.

Outcome 3 — Scalability Under Heterogeneity FedEx-LoRA accuracy should remain stable as client count scales from 10 to 50. FedAvg-LoRA divergence is expected to grow with k , confirming the compounding nature of aggregation bias in large-scale deployments.

Outcome 4 — Cross-Task Generalisation On MetaMathQA/GSM8K and E2E NLG, FedEx-LoRA should match centralised LoRA within 2%, while FedAvg-LoRA shows 3–6% degradation, validating that exact aggregation generalises across reasoning and generation tasks.

Outcome 5 — SVD Compression Pareto Frontier Residual compression at $r' = r/2$ is expected to achieve less than 1.5% accuracy loss on GLUE SST-2 while reducing per-round communication by approximately 50%, providing a practical deployment guideline for bandwidth-constrained networks.

Outcome 6 — Synchronisation Guidelines Increasing local epochs E is expected to amplify $\|\Delta W_{\text{res}}\|_F$, confirming that more local training increases aggregation complexity. Optimal E is expected to fall in the range $\{3, 5\}$ for most benchmarks, contributing empirical synchronisation guidelines.

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